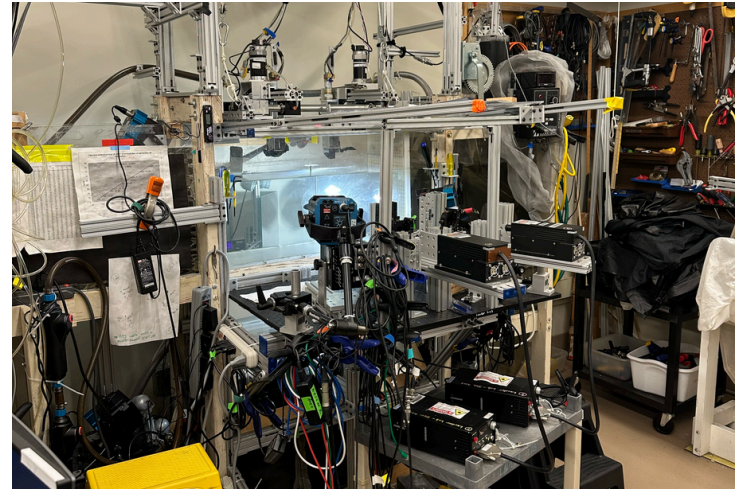


FISH ROBOT SWIMMING EFFICIENCY - HARVARD UNIVERSITY



What?

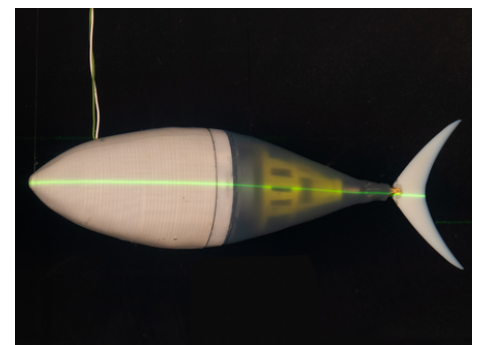
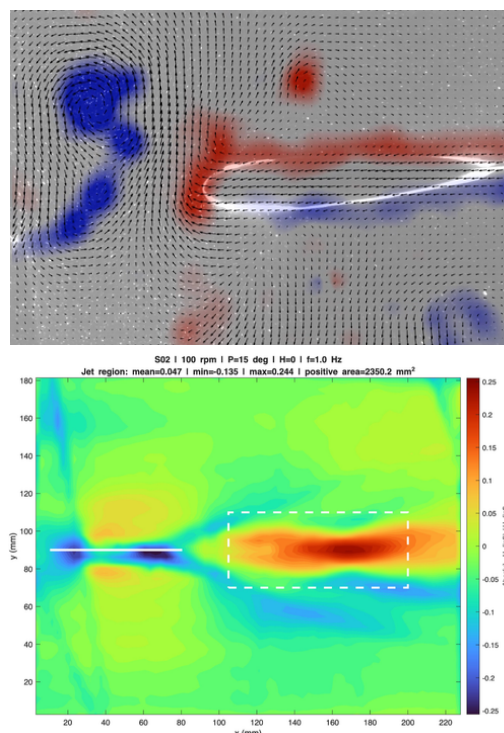
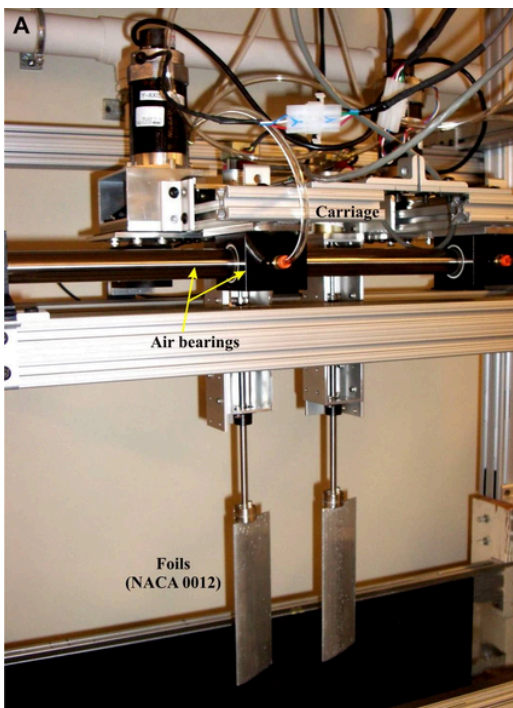
- Investigate how an actuated hydrofoil and a passive 3-D-printed fish swim and hold station in **thrust-producing vortex wakes**.
- Study how frequency, pitch amplitude, and flow speed affect swimming and station-holding performance.

How?

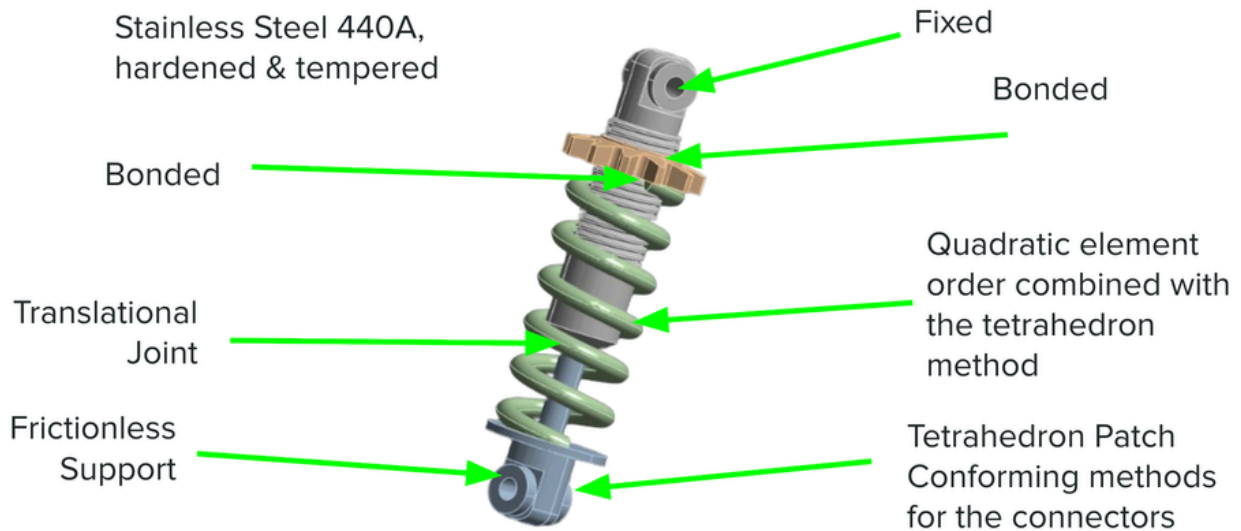
- Operated and redesigned a robotic flapper with air bearings, **LabVIEW**, **PID control**, and **NI-DAQ**.
- Ran hydrofoil and passive-fish experiments across flow conditions.
- Performed **particle image velocimetry (PIV)**; processed wake data in **MATLAB** and PIVlab.
- Built 2-D tracking and 3-D reconstruction pipelines in **MATLAB** and **R**.

Results

- Presented findings at the **ONR MURI** Annual Review Meeting (Sep 2026) and the **APS Division of Fluid Dynamics Meeting** (Nov 2026).
- Quantified swimming kinematics across flow conditions, that can be used in ship propulsion and underwater drones group design.



SHOCK ABSORBER - FEA MODAL ANALYSIS



What?

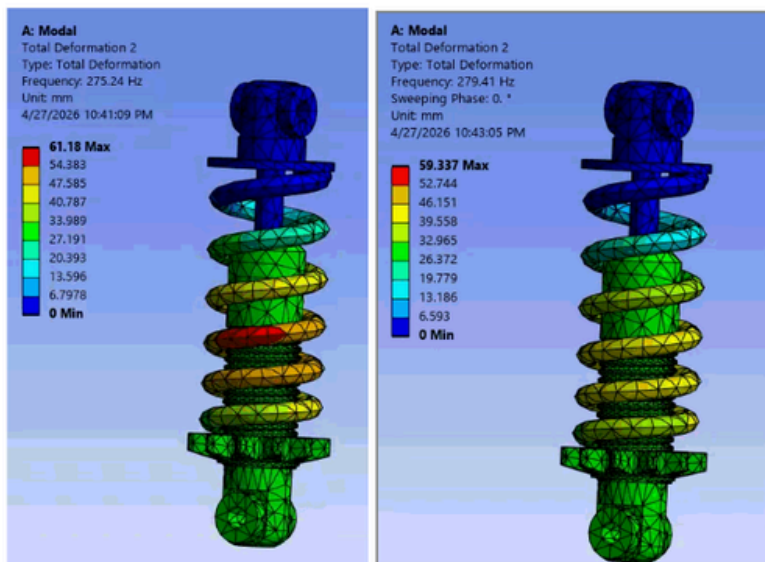
- Performed modal analysis on a shock absorber assembly to study its natural vibration behavior.
- Studied the natural vibration behavior of a scaled shock absorber assembly.
- Tested how damping affects vibration decay and stability.

How?

- Set up the model in ANSYS with fixed, bonded, frictionless, and translational joint boundary conditions.
- Used Stainless Steel 440A and quadratic tetrahedral mesh elements.
- Tested damping ratios from 0.05 to 0.3.

Results

- Natural frequencies stayed around 152–712 Hz.
- Higher damping caused faster vibration decay without major frequency shifts.
- Results showed the system is mainly mass–stiffness dominated.
- Fixed contact issues by improving boundary conditions and using a translational joint where sliding motion was needed.



Damping off

Damping on

0.05 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	151.89	-1.551	1.0211e-002	-6.4161e-002
2. 2.	355.23	-5.3157	1.4963e-002	-9.4023e-002
3. 3.	367.36	-5.5009	1.4994e-002	-9.4222e-002
4. 4.	414.7	-6.1958	1.4939e-002	-9.3874e-002
5. 5.	419.73	-6.2565	1.4904e-002	-9.3658e-002
6. 6.	704.77	-10.564	1.4988e-002	-9.4183e-002

0.1 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	152.02	-5.1625	3.3939e-002	-0.21337
2. 2.	355.63	-17.699	4.9706e-002	-0.3127
3. 3.	367.77	-18.342	4.9811e-002	-0.31336
4. 4.	415.17	-20.629	4.9627e-002	-0.3122
5. 5.	420.2	-20.832	4.9515e-002	-0.31149
6. 6.	705.59	-35.176	4.9791e-002	-0.31323

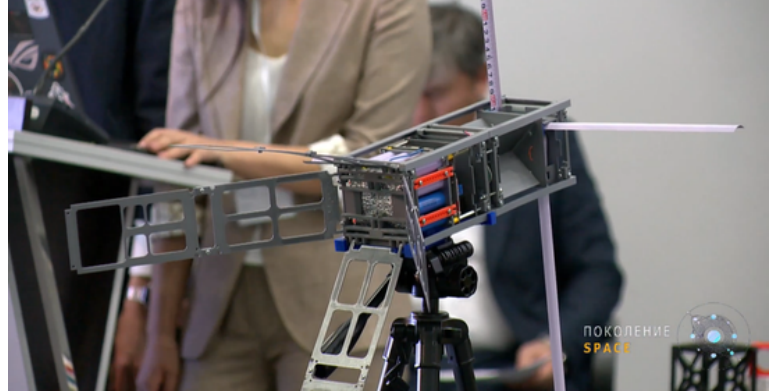
0.2 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	152.46	-10.276	6.7251e-002	-0.42351
2. 2.	356.94	-35.268	9.8328e-002	-0.62082
3. 3.	369.13	-36.549	9.8533e-002	-0.62213
4. 4.	416.7	-41.106	9.817e-002	-0.61981
5. 5.	421.73	-41.512	9.7959e-002	-0.61847
6. 6.	708.23	-70.097	9.8482e-002	-0.62187

0.3 damping ratio

Mode	Damped Frequency [Hz]	Stability [Hz]	Modal Damping Ratio	Logarithmic Decrement
1. 1.	153.16	-15.297	9.938e-002	-0.62753
2. 2.	359.06	-52.587	0.14491	-0.92021
3. 3.	371.34	-54.498	0.14521	-0.92214
4. 4.	419.19	-61.231	0.14467	-0.91867
5. 5.	424.22	-61.901	0.14439	-0.91682
6. 6.	712.5	-104.52	0.14515	-0.92175

SKILLSAT STARTUP



What?

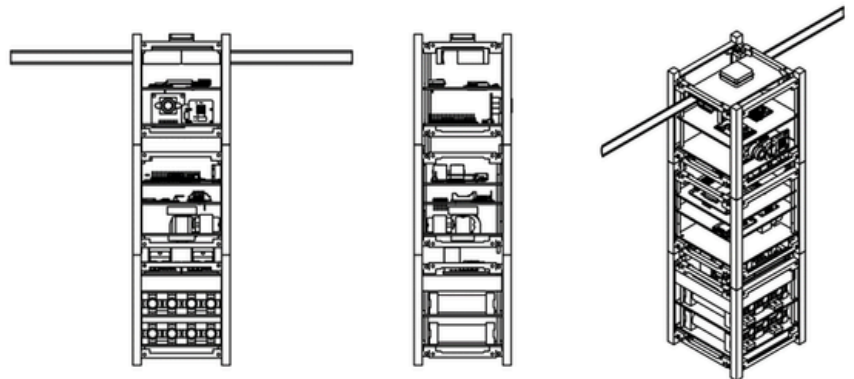
- Founded SkillSat, an EdTech startup to manufacture STEM kits that teach students to build stratospheric satellites.

How?

- Used Arduino and Raspberry Pi as the base for satellite systems.
- Utilized additive manufacturing to prototype 25+ designs.
- Applied DFA and DFM principles to reduce production costs by 44%.
- Tested prototypes at the National Space Center of Kazakhstan.

Results

- Launched satellites to Near Space (40 km+) via stratospheric balloons.
- Raised \$15K in funding and distributed 80+ kits to 240+ students across 25+ schools.
- Recognized as a Top-3 SpaceTech and Top-15 EdTech Startup by Forbes Kazakhstan.



ABYLAIKHAN MUKHAMEJANOV

MECHANICAL ENGINEERING AT THE LEHIGH UNIVERSITY

✉ abylaikhan.mukhamejanov@gmail.com

in [linkedin.com/in/abylaikhan-m/](https://www.linkedin.com/in/abylaikhan-m/)

☎ +1 (484) 353-9865



Uncovering the Fluid-Structure Interactions of Wave-Assisted Propulsion via Cyber-Physical Fluid Dynamics



Abylaikhan Mukhamejanov¹, Ata Tankut Ardic¹, Keith W. Moored¹

¹Department of Mechanical Engineering & Mechanics, Lehigh University, Bethlehem, PA 18015

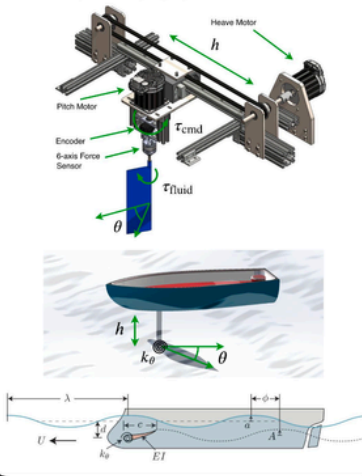
78th Annual Meeting APS DFD
Nov. 23–25, 2025, Houston, TX

Project Objectives and Goals

- Investigate wave-assisted propulsion (WAP) using controlled heave and passive pitch to transform wave energy into forward thrust and improve stability.
- Develop and validate a cyber-physical testbed that couples a flapping hydrofoil with virtual mass-spring-damper dynamics.
- Implement real-time control of virtual stiffness, damping, and inertia to emulate passive pitching behavior.

Motivation

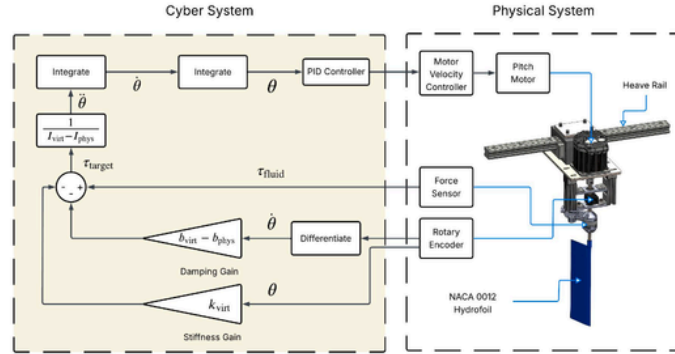
- Wave-assisted propulsion (WAP) uses oscillating hydrofoils to extract energy from waves and generate forward thrust.
- The flapping motion can also reduce vessel oscillations, improving stability and efficiency in marine systems.
- Studying these fluid-structure interactions (FSI) requires precise control of mass, stiffness, and damping parameters.
- Traditional experiments fix mass, stiffness, and damping through mechanical design, limiting flexibility in studying dynamic behavior.



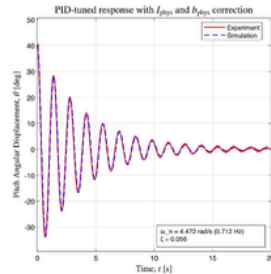
Methods & Procedures

- A cyber-physical system [1] couples the physical actuator-fluid testbed with a computer-based controller that enforces a virtual dynamic model in real time.
- Command signal cancels physical inertia and damping to match desired virtual parameters.
- Physical inertia and damping identified via ring-down experiment.
- Double integration [2] of force/torque removes filtering need and enables stable 4 kHz Simulink Real-Time control loop.

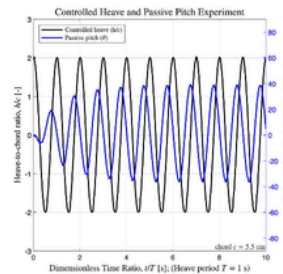
$$\begin{aligned}\tau &= I \ddot{\theta} + b \dot{\theta} + k \theta \\ \tau_{\text{virt}} &= -(I_v \ddot{\theta} + b_v \dot{\theta} + k_v \theta) \\ I_p \ddot{\theta} &= \tau_{\text{fluid}} + \tau_{\text{virt}} \\ (I_p + I_v) \ddot{\theta} + (b_p + b_v) \dot{\theta} + k_v \theta &= \tau_{\text{fluid}}\end{aligned}$$



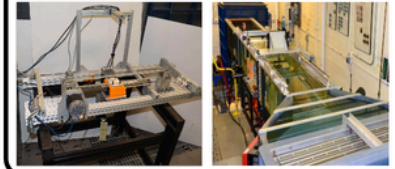
System Validation



Air Experiment Results



Experimental Setup



Conclusions

- Developed a working cyber-physical system mimicking passive pitching.
- Any structural parameters can be virtually assigned to the system, including nonlinear stiffness elements and damping characteristics.
- The system accepts real heave motion inputs from naval vessels, enabling experimental investigation and optimization of Wave-Assisted Propulsion (WAP).

Future Studies / Recommendations

- Evaluate propulsion efficiency and stability under varying environmental and structural parameters in the water channel.
- Recreate natural sea-state conditions by coupling heave-wave excitation with vessel motions across multiple degrees of freedom.

Acknowledgments

This research was supported by the Lehigh Opportunity Fund and Rossin Research Scholars Program (2025), managed by Prof. Mark Snyder, Associate Dean for Graduate Education, Lehigh University.

References

- [1] Mackowski, A. W., and Williamson, C. H. K., 2011, "Developing a Cyber-Physical Fluid Dynamics Facility for Fluid-Structure Interaction Studies," *J. Fluids Struct.*, 27 (5–6), pp. 748–757.
- [2] Zhu, Y., Su, Y., and Breat, K., 2020, "Nonlinear Flow-Induced Instability of an Elastically Mounted Pitching Wing," *J. Fluid Mech.*, 899, A35.